USN | | | | | | | | | | [23CSBS302]

THIRD SEMESTER B.Tech EXAMINATION, DECEMBER - 2024

COMPUTER ORGANIZATION & ARCHITECTURE

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

			Blooms Level	CO's	Marks
		PART-A			**5 x2=10**
1.	**a.**	What are the types of operations performed by instructions?	L1	CO1	**02**
	b.	What is underflow in floating point arithmetic?	L1	CO2	**02**
	c.	What are the advantages and disadvantages of micro programmed control Unit?	L1	CO3	**02**
	d.	What is meant by Data Transfer?	L2	CO4	**02**
	e.	In memory organization, what is temporal locality?	L1	CO5	**02**
		PART-B			**6 X 5=30**
2.	**a.**	Explain different types of interrupts in computer system.	L2	CO1	**05**
		OR			
	b.	Explain zero address instruction formats.	L2	CO1	**05**
3.	**a.**	State and explain the arithmetic multiplication operations on floating point numbers.	L2	CO2	**05**
		OR			
	b.	State and explain the arithmetic addition operations on floating point numbers.	L2	CO2	**05**
4.	**a.**	Discuss about Hardwired Control unit in detail.	L2	CO3	**05**
		OR			
	b.	Discuss about Micro Programmed Control unit in detail.	L2	CO3	**05**
5.	**a.**	Explain different types of data transfer instructions.	L2	CO4	**05**
		OR			
	b.	Explain different types of interrupts in computer system	L2	CO4	**05**
6.	**a.**	Considering memory hierarchy, define hit and miss.	L2	CO5	**05**
		OR			
	b.	Explain the purpose of **i.** cache memory **ii.** auxiliary memory	L2	CO5	**05**

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7.	**a.**	Discuss the disadvantage of Flash memory.	L2	CO5	**05**
		OR			
	b.	Discuss briefly about semiconductor RAM memory.	L2	CO5	**05**
		PART-C			**6 X 10=60**
8.	**a.**	Discuss about the execution of a complete instruction.	L2	CO1	**10**
		OR			
	b.	Enumerate the most commonly used addressing modes of CPU instructions.	L2	CO1	**10**
9.	**a.**	With an example discuss about Booth multiplication algorithm.	L2	CO2	**10**
		OR			
	b.	State the purpose of Look Ahead Carry Adder. Derive the expressions for propagate and generate functions of a 4-bit Look Ahead Carry Adder and draw its schematic. .	L2	CO2	**10**
10.	**a.**	What is the purpose of parallel processing? Discuss about different forms of parallel processing.	L1	CO3	**10**
		OR			
	b.	List the types of hazards and briefly explain the impact of such hazards on the performance of the pipelines.	L1	CO3	**10**
11.	**a.**	Describe in detail about DMA operations with neat diagram.	L2	CO4	**10**
		OR			
	b.	Explain the different types of I/O communication techniques. Which is the most efficient one?	L2	CO4	**10**
12.	**a.**	With a neat diagram, explain the memory hierarchy.	L2	CO5	**10**
		OR			
	b.	Explain the use of Translation Lookaside Buffers in Virtual Memory organization. With a neat sketch explain the organization of associative-mapped TLB.	L2	CO5	**10**
13.	**a.**	What is cache memory? How to improve cache performance? Explain.	L1	CO5	**10**
		OR			
	b.	What is associative memory? Why is it faster than the main memory? Explain its hardware organization in detail.	L1	CO5	**10**